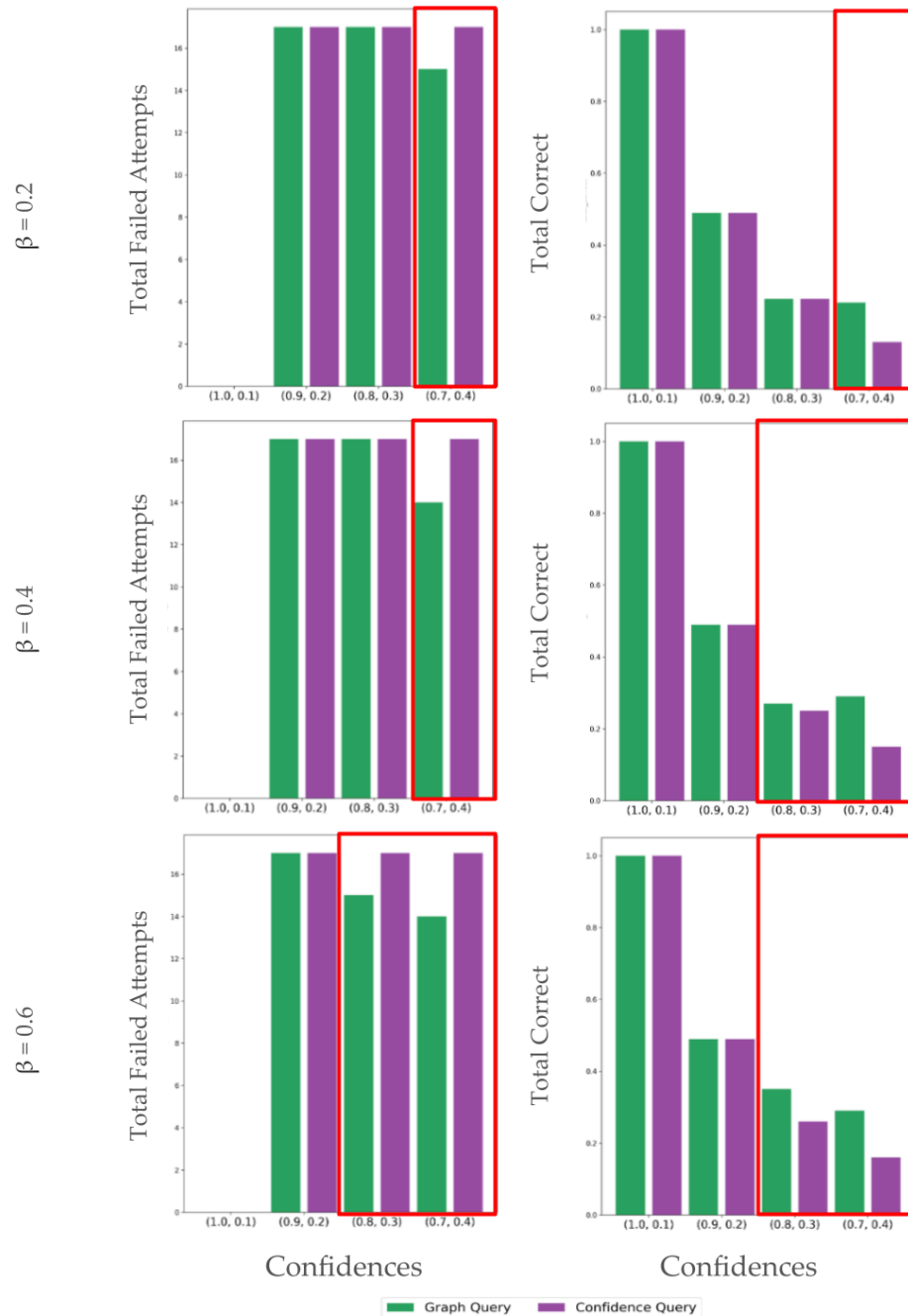


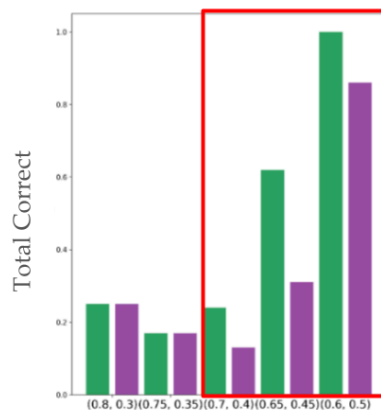
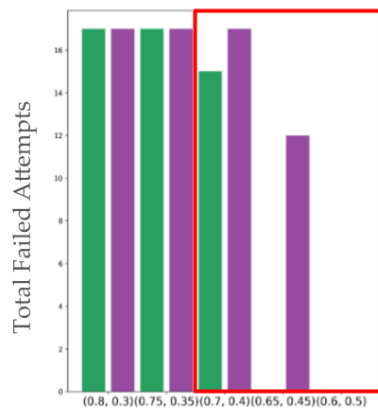
Rebuttal Supplementary Materials

GraphQuery v. ConfidenceQuery: Simulation Results

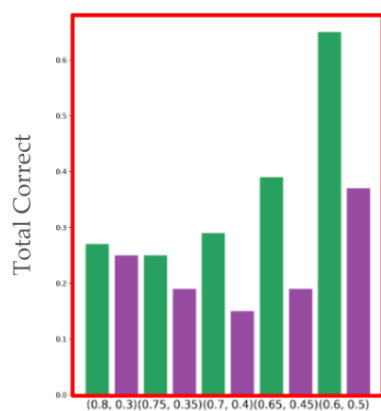
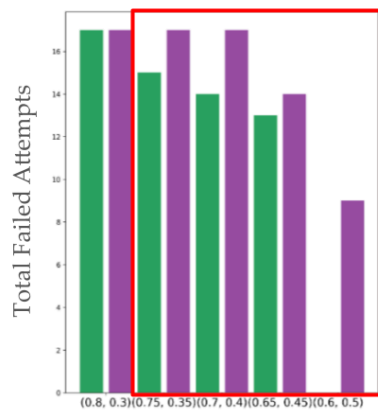
Uniform random noise in each module's query cost q_i
 $q_i \sim \text{Uniform}[(1-\beta)Q, (1+\beta)Q]$, $Q = 0.32$



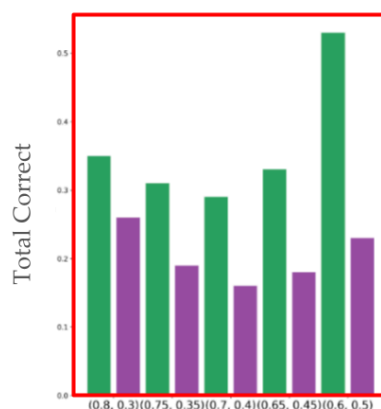
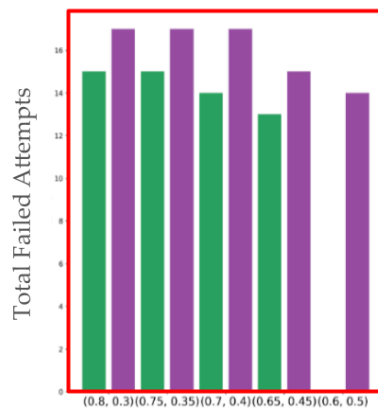
$\beta = 0.2$



$\beta = 0.4$

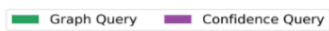


$\beta = 0.6$



Confidences

Confidences



GraphQuery v. ConfidenceQuery: Feeding Study Results

Subjective scores (top) and objective scores (bottom) for N=5 users without mobility limitations.

Means	Mental/Physical Demand ↓	Effort ↓	Subjective Success ↑	Satisfaction ↑
GraphQuery	1.8	2	4.2	3.8
ConfidentQuery	3.4	3.6	2.2	1.6
Means	Mean Queries per Bite ↓	Mean Executions per Bite ↓	Mean Successful Bites per Trial ↑	
GraphQuery	2.32	1.68	0.92	
ConfidentQuery	6.28	1.12	0.44	

System Recommendations

We show how our design insights generalize by considering algorithm recommendations for two other modular systems. To illustrate this, we consider another feeding policy architecture with visuo-haptic perceptual redundancy [2], as well as a large multi-robot swarm setting [3]. We would model the first architecture with an OR-then-AND redundancy structure and lower query cost of $q_i = 0.16$. This leads to the same recommendation as in our setting (based on Fig. 4). For the multi-robot swarm domain, we would select $N=100$ to approximate the order-of-magnitude in the system, and use the OR redundancy structure. We would now recommend only BruteForce due to its superior scalability (based on Fig. 4, first column), while still recommending either Query-until-Confident variant due to their superior performance in the Total Timesteps and Total Correct metrics.

References

- [1] R. Banerjee, R.K. Jenamani, S. Vasudev, A. Nanavati, S. Dean, and T. Bhattacharjee. 2024. To Ask or Not To Ask: Human-in-the-loop Contextual Bandits with Applications in Robot-Assisted Feeding, in IEEE International Conference on Robotics and Automation (ICRA), 2025.
- [2] Z. Wu, B. Ai, T. Silver, and T. Bhattacharjee. 2025. SAVOR: Skill Affordance Learning from Visuo-Haptic Perception for Robot-Assisted Bite Acquisition, in Conference on Robot Learning (CoRL), 2025.
- [3] M. Rubenstein., A. Cornejo., and R. Nagpal, 2014. Programmable self-assembly in a thousand-robot swarm. Science, 345(6198), pp.795-799.